Laboratorio de contenedores Docker

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Cloud 2 Betek

Laboratorio de contenedores Docker

0) Preparación

Verifica Docker:

```
andrea@Andrea:~$ docker version
Client:
 Version:           29.3.1
 API version:       1.54
 Go version:        go1.25.8
 Git commit:        c2be9cc
 Built:             Wed Mar 25 16:12:39 2026
 OS/Arch:           linux/amd64
 Context:           default

Server: Docker Desktop 4.67.0 (222858)
 Engine:
  Version:          29.3.1
  API version:      1.54 (minimum version 1.40)
  Go version:       go1.25.8
  Git commit:       f78c987
  Built:            Wed Mar 25 16:13:48 2026
  OS/Arch:          linux/amd64
  Experimental:     false
 containerd:
  Version:          v2.2.1
  GitCommit:        dea7da592f5d1d2b7755e3a161be07f43fad8f75
 runc:
  Version:          1.3.4
  GitCommit:        v1.3.4-0-gd6d73eb8
 docker-init:
  Version:          0.19.0
  GitCommit:        de40ad0
```

docker info:

```
andrea@Andrea:~$ docker info
Client:
 Version:    29.3.1
 Context:    default
 Debug Mode: false
 Plugins:
  agent: Docker AI Agent Runner (Docker Inc.)
    Version:  v1.34.0
    Path:     /usr/local/lib/docker/cli-plugins/docker-agent
  ai: Docker AI Agent - Ask Gordon (Docker Inc.)
    Version:  v1.20.1
    Path:     /usr/local/lib/docker/cli-plugins/docker-ai
  buildx: Docker Buildx (Docker Inc.)
    Version:  v0.32.1-desktop.1
    Path:     /usr/local/lib/docker/cli-plugins/docker-buildx
  compose: Docker Compose (Docker Inc.)
    Version:  v5.1.1
    Path:     /usr/local/lib/docker/cli-plugins/docker-compose
  debug: Get a shell into any image or container (Docker Inc.)
    Version:  0.0.47
    Path:     /usr/local/lib/docker/cli-plugins/docker-debug
  desktop: Docker Desktop commands (Docker Inc.)
    Version:  v0.3.0
    Path:     /usr/local/lib/docker/cli-plugins/docker-desktop
  dhi: CLI for managing Docker Hardened Images (Docker Inc.)
    Version:  v0.0.2
    Path:     /usr/local/lib/docker/cli-plugins/docker-dhi
  extension: Manages Docker extensions (Docker Inc.)
    Version:  v0.2.31
    Path:     /usr/local/lib/docker/cli-plugins/docker-extension
  init: Creates Docker-related starter files for your project (Docker Inc.)
    Version:  v1.4.0
    Path:     /usr/local/lib/docker/cli-plugins/docker-init
  mcp: Docker MCP Plugin (Docker Inc.)
    Version:  v0.40.3
    Path:     /usr/local/lib/docker/cli-plugins/docker-mcp
  model: Docker Model Runner (Docker Inc.)
    Version:  v1.1.25
```

```
Server:
 Containers: 0
  Running: 0
  Paused: 0
  Stopped: 0
 Images: 5
 Server Version: 29.3.1
 Storage Driver: overlayfs
  driver-type: io.containerd.snapshotter.v1
 Logging Driver: json-file
 Cgroup Driver: cgroupfs
 Cgroup Version: 2
 Plugins:
  Volume: local
  Network: bridge host ipvlan macvlan null overlay
  Log: awslogs fluentd gcplogs gelf journald json-file local splunk syslog
 CDI spec directories:
  /etc/cdi
  /var/run/cdi
 Discovered Devices:
  cdi: docker.com/gpu=webgpu
 Swarm: inactive
 Runtimes: nvidia runc io.containerd.runc.v2
 Default Runtime: runc
 Init Binary: docker-init
 containerd version: dea7da592f5d1d2b7755e3a161be07f43fad8f75
 runc version: v1.3.4-0-gd6d73eb8
 init version: de40ad0
 Security Options:
  seccomp
   Profile: builtin
  cgroupns
 Kernel Version: 6.6.87.2-microsoft-standard-WSL2
 Operating System: Docker Desktop
 OSType: linux
 Architecture: x86_64
 CPUs: 12
 Total Memory: 5.708GiB
```

1) Docker Image

1.1 Descargar una imagen

```
andrea@Andrea:~$ docker pull nginx:latest
latest: Pulling from library/nginx
Digest: sha256:7150b3a39203cb5bee612ff4a9d18774f8c7caf6399d6e8985e97e28eb751c18
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
andrea@Andrea:~$ docker images

IMAGE                    ID             DISK USAGE   CONTENT SIZE   EXTRA
alpine:latest            25109184c71b       13.1MB         3.95MB
mi-sitio-web:latest      12226e992320        237MB           63MB
miapp:latest             d230c7f673fa       12.1MB         3.63MB
miapp:v1                 d230c7f673fa       12.1MB         3.63MB
nginx:latest             7150b3a39203        240MB         65.8MB
polinux/stress:latest    b6144f84f9c1         20MB         4.04MB
andrea@Andrea:~$
```

1.2 Construir una imagen desde Dockerfile

```
andrea@Andrea:~$ cd docker-lab
andrea@Andrea:~/docker-lab$ cat <<EOF > Dockerfile
FROM alpine:3.20
WORKDIR /app
RUN echo "Hola desde Docker" > /app/mensaje.txt
CMD ["cat", "/app/mensaje.txt"]
EOF
andrea@Andrea:~/docker-lab$ docker build -t miapp:v1 .
[+] Building 1.3s (8/8) FINISHED                                                                                    docker:default
 => [internal] load build definition from Dockerfile                                                                          0.0s
 => => transferring dockerfile: 147B                                                                                          0.0s
 => [internal] load metadata for docker.io/library/alpine:3.20                                                                1.0s
 => [auth] library/alpine:pull token for registry-1.docker.io                                                                 0.0s
 => [internal] load .dockerignore                                                                                             0.0s
 => => transferring context: 2B                                                                                               0.0s
 => [1/3] FROM docker.io/library/alpine:3.20@sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03805          0.0s
 => => resolve docker.io/library/alpine:3.20@sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03805          0.0s
 => CACHED [2/3] WORKDIR /app                                                                                                 0.0s
 => CACHED [3/3] RUN echo "Hola desde Docker" > /app/mensaje.txt                                                              0.0s
 => exporting to image                                                                                                        0.1s
 => => exporting layers                                                                                                       0.0s
 => => exporting manifest sha256:5f1974616ba320ecb41b41ef4140dddc4d08ff5c00f8ee4adfeb6e31479abf4d                             0.0s
 => => exporting config sha256:9284d0e3d899836acda8072ca99bbdc6c48c5561021aa9b549a2d2bdacd0cf6b                               0.0s
 => => exporting attestation manifest sha256:a5662440d63f9f9c516b7a95a51513a50e76c42c339b043b5b226f66cfcc2a0f                  0.1s
 => => exporting manifest list sha256:84e7c5ab379588f246591fb6174bb55ded1486715385735054bf8cb1a28d4b31                        0.0s
 => => naming to docker.io/library/miapp:v1                                                                                   0.0s
 => => unpacking to docker.io/library/miapp:v1                                                                                0.0s
andrea@Andrea:~/docker-lab$
```

Comprobar:

```
andrea@Andrea:~/docker-lab$ docker images | grep miapp
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
miapp:latest            d230c7f673fa       12.1MB         3.63MB
miapp:v1                84e7c5ab3795       12.1MB         3.63MB
andrea@Andrea:~/docker-lab$
```

Ejecuta para validar:

```
andrea@Andrea:~/docker-lab$ docker run --rm miapp:v1
Hola desde Docker
andrea@Andrea:~/docker-lab$
```

1.3 Etiquetar una imagen

```
andrea@Andrea:~/docker-lab$ docker tag miapp:v1 miapp:latest
andrea@Andrea:~/docker-lab$ docker image inspect miapp:v1 --format '{{.Id}}'
sha256:84e7c5ab379588f246591fb6174bb55ded1486715385735054bf8cb1a28d4b31
andrea@Andrea:~/docker-lab$
```

1.4 Guardar y Cargar (Backup)

Esto simula cómo moverías una imagen sin usar un registro:

```
andrea@Andrea:~/docker-lab$ docker save -o miapp.tar miapp:v1
andrea@Andrea:~/docker-lab$ ls -lh miapp.tar
-rw------- 1 andrea andrea 3.5M Apr  5 21:08 miapp.tar
andrea@Andrea:~/docker-lab$
```

Borrar y recuperar:

```
andrea@Andrea:~/docker-lab$ docker rmi miapp:v1
Untagged: miapp:v1
andrea@Andrea:~/docker-lab$ docker load -i miapp.tar
Loaded image: miapp:v1
andrea@Andrea:~/docker-lab$ docker images | grep miapp
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
miapp:latest            84e7c5ab3795       12.1MB         3.63MB
miapp:v1                84e7c5ab3795       12.1MB         3.63MB
andrea@Andrea:~/docker-lab$
```

2) Docker Container

2.1 Ejecutar un contenedor

```
andrea@Andrea:~/docker-lab$ docker run -d --name web1 nginx:latest
7589feda937a2b527c9e77cfb315247a598b60bcc6e84c0539fc783e1f354259
andrea@Andrea:~/docker-lab$
```

Comprobación:

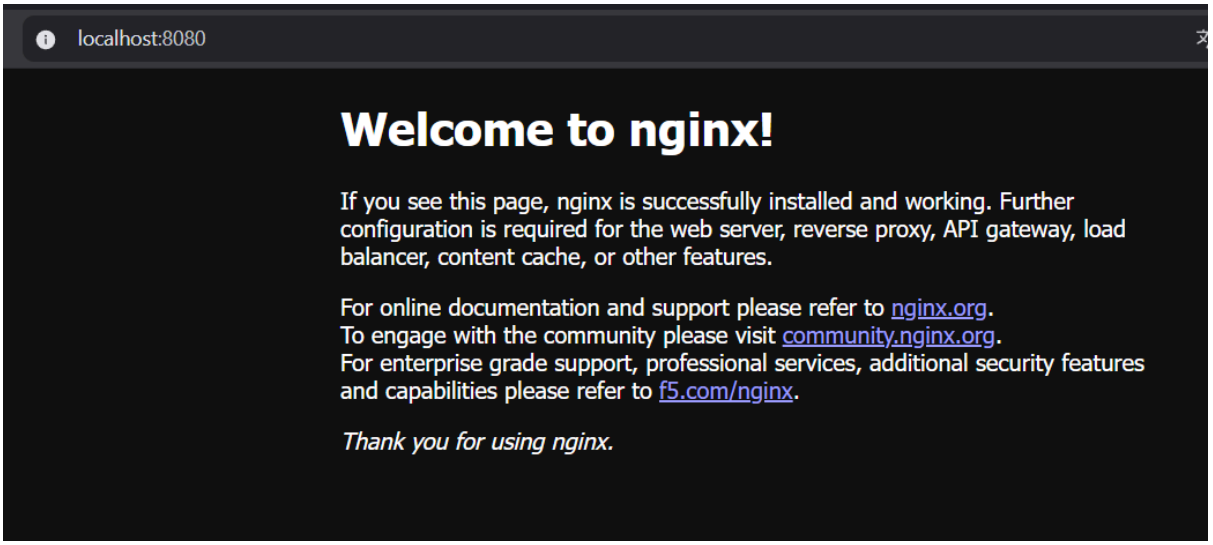
```
andrea@Andrea:~/docker-lab$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS     NAMES
7589feda937a   nginx:latest   "/docker-entrypoint.…"   3 minutes ago    Up 3 minutes    80/tcp    web1
```

2.2 Ejecutar contenedor con nombre y puertos

```
andrea@Andrea:~/docker-lab$ docker run -d --name web2 -p 8080:80 nginx:latest
7b9615c2bea8d06c0f3db2b33245fee93381a66c2a3738cf4d467a9dff4c510a
```

Comprobación:

```
andrea@Andrea:~/docker-lab$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED              STATUS              PORTS                                       NAMES
7b9615c2bea8   nginx:latest   "/docker-entrypoint.…"   About a minute ago   Up About a minute   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp     web2
7589feda937a   nginx:latest   "/docker-entrypoint.…"   6 minutes ago        Up 6 minutes        80/tcp                                      web1
andrea@Andrea:~/docker-lab$
```



2.3 al 2.5 Administración y Logs:

Listar todo:

```
andrea@Andrea:~/docker-lab$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS                                       NAMES
7b9615c2bea8   nginx:latest   "/docker-entrypoint.…"   5 minutes ago    Up 5 minutes    0.0.0.0:8080->80/tcp, [::]:8080->80/tcp     web2
7589feda937a   nginx:latest   "/docker-entrypoint.…"   10 minutes ago   Up 10 minutes   80/tcp                                      web1
```

Ver qué pasa dentro:

```
andrea@Andrea:~/docker-lab$ docker logs web2
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/04/06 02:22:39 [notice] 1#1: using the "epoll" event method
2026/04/06 02:22:39 [notice] 1#1: nginx/1.29.7
2026/04/06 02:22:39 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/04/06 02:22:39 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/04/06 02:22:39 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/04/06 02:22:39 [notice] 1#1: start worker processes
2026/04/06 02:22:39 [notice] 1#1: start worker process 29
2026/04/06 02:22:39 [notice] 1#1: start worker process 30
2026/04/06 02:22:39 [notice] 1#1: start worker process 31
2026/04/06 02:22:39 [notice] 1#1: start worker process 32
2026/04/06 02:22:39 [notice] 1#1: start worker process 33
2026/04/06 02:22:39 [notice] 1#1: start worker process 34
2026/04/06 02:22:39 [notice] 1#1: start worker process 35
2026/04/06 02:22:39 [notice] 1#1: start worker process 36
2026/04/06 02:22:39 [notice] 1#1: start worker process 37
2026/04/06 02:22:39 [notice] 1#1: start worker process 38
2026/04/06 02:22:39 [notice] 1#1: start worker process 39
2026/04/06 02:22:39 [notice] 1#1: start worker process 40
172.17.0.1 - - [06/Apr/2026:02:25:46 +0000] "GET / HTTP/1.1" 200 896 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, l
ike Gecko) Chrome/146.0.0.0 Safari/537.36" "-"
172.17.0.1 - - [06/Apr/2026:02:25:46 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://localhost:8080/" "Mozilla/5.0 (Windows NT 10.0; Win64; x
64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36" "-"
2026/04/06 02:25:46 [error] 29#29: *1 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 172.17.0.1, serv
er: localhost, request: "GET /favicon.ico HTTP/1.1", host: "localhost:8080", referrer: "http://localhost:8080/"
```

Inspección rápida:

```
andrea@Andrea:~/docker-lab$ docker inspect web2 --format '{{.State.Status}}
running
```

2.6 y 2.7 Ciclo de Vida y Modo Interactivo

Detener y Arrancar:

```
andrea@Andrea:~/docker-lab$ docker stop web2
web2
andrea@Andrea:~/docker-lab$ docker start web2
web2
```

Entrar al contenedor:

```
andrea@Andrea:~/docker-lab$ docker run -it --name alpine1 alpine:3.20 sh
Unable to find image 'alpine:3.20' locally
3.20: Pulling from library/alpine
Digest: sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03805
Status: Downloaded newer image for alpine:3.20
/ #
```

Dentro del contenedor

```
/ # cat /etc/os-release
NAME="Alpine Linux"
ID=alpine
VERSION_ID=3.20.9
PRETTY_NAME="Alpine Linux v3.20"
HOME_URL="https://alpinelinux.org/"
BUG_REPORT_URL="https://gitlab.alpinelinux.org/alpine/aports/-/issues"
/ #
```

2.9 Limpieza de Contenedores

Para uno detenido:

```
andrea@Andrea:~/docker-lab$ docker rm alpine1
alpine1
```

Para forzar el borrado de uno activo:

```
andrea@Andrea:~/docker-lab$ docker rm -f web2
web2
```

Validar:

```
andrea@Andrea:~/docker-lab$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS     NAMES
7589feda937a   nginx:latest   "/docker-entrypoint.…"   20 minutes ago   Up 20 minutes   80/tcp    web1
```

3) Volúmenes

3.1 Crear volumen

```
andrea@Andrea:~/docker-lab$ docker volume create datos_nginx
datos_nginx
andrea@Andrea:~/docker-lab$ docker volume ls
DRIVER    VOLUME NAME
local     datos_nginx
```

3.2 Inspeccionar volumen

```
andrea@Andrea:~/docker-lab$ docker volume inspect datos_nginx
[
    {
        "CreatedAt": "2026-04-06T02:40:10Z",
        "Driver": "local",
        "Labels": null,
        "Mountpoint": "/var/lib/docker/volumes/datos_nginx/_data",
        "Name": "datos_nginx",
        "Options": null,
        "Scope": "local"
    }
]
```

3.3 Usar el Volumen en un Contenedor

Lanzar el contenedor:

```
andrea@Andrea:~/docker-lab$ docker run -d --name webvol -p 8081:80 -v datos_nginx:/usr/share/nginx/html nginx:latest
0b77c29b6d742ccbf31b941b6563ff363c786f04c583c9024466d9006fe2bbc6
```

Crear un archivo local en Ubuntu y copiar el archivo al contenedor:

```
andrea@Andrea:~/docker-lab$ echo "<h1>Hola volumen</h1>" > index.html
andrea@Andrea:~/docker-lab$ docker cp index.html webvol:/usr/share/nginx/html/index.html
Successfully copied 2.05kB to webvol:/usr/share/nginx/html/index.html
```

Comprobación:

Hola volumen

3.4 Eliminar volumen

Borra el contenedor:

```
andrea@Andrea:~/docker-lab$ docker rm -f webvol
webvol
```

Crea uno nuevo usando el mismo volumen:

```
andrea@Andrea:~/docker-lab$ docker run -d --name webvol2 -p 8082:80 -v datos_nginx:/usr/share/nginx/html nginx:latest
7506f16caa9abf6e47f36d5201c1fafef14a614d1dc5d993f0ef724c1d99be2a
```

Verificación:

Hola volumen

Limpieza Final

```
andrea@Andrea:~/docker-lab$ docker rm -f webvol2
webvol2
andrea@Andrea:~/docker-lab$ docker volume rm datos_nginx
datos_nginx
```

4) Redes

4.1 Listar redes

```
andrea@Andrea:~/docker-lab$ docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
715053ebe745   bridge    bridge    local
cb7a0f91b259   host      host      local
81500d894020   none      null      local
```

4.2 Crear red bridge propia

```
andrea@Andrea:~/docker-lab$ docker network create mi_red
1b201871505f7add358228c3d64b5f6fce7b90a3fe74bd3af2ec2a1d5804b5f1
```

Comprobación:

```
andrea@Andrea:~/docker-lab$ docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
715053ebe745   bridge    bridge    local
cb7a0f91b259   host      host      local
1b201871505f   mi_red    bridge    local
81500d894020   none      null      local
```

4.3 Inspeccionar red

```
andrea@Andrea:~/docker-lab$ docker network inspect mi_red
[
    {
        "Name": "mi_red",
        "Id": "1b201871505f7add358228c3d64b5f6fce7b90a3fe74bd3af2ec2a1d5804b5f1",
        "Created": "2026-04-06T02:56:11.661885961Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "172.18.0.0/16",
                    "Gateway": "172.18.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {
            "com.docker.network.enable_ipv4": "true",
            "com.docker.network.enable_ipv6": "false"
        },
        "Labels": {},
        "Containers": {},
        "Status": {
            "IPAM": {
                "Subnets": {
                    "172.18.0.0/16": {
                        "IPsInUse": 3,
                        "DynamicIPsAvailable": 65533
                    }
                }
            }
        }
    }
]
```

4.4 Conectar Contenedores y Probar Comunicación

Lanzar el primer contenedor:

```
andrea@Andrea:~/docker-lab$ docker run -d --name caja1 --network mi_red alpine:3.20 sleep 300
2b3c0f8b3b25ea31394347dbd4f6d2506c4332b8cb5e3bb4cabcae8091eb09f1
```

Lanzar el segundo contenedor:

```
andrea@Andrea:~/docker-lab$ docker run -d --name caja2 --network mi_red alpine:3.20 sleep 300
87a03dfe78daa0d8513784fdd0f446db9c78611971a63dffc37cf79a81baeddc
```

Comprobación de conectividad por nombre:

```
andrea@Andrea:~/docker-lab$ docker exec caja1 ping -c 2 caja2
PING caja2 (172.18.0.3): 56 data bytes
64 bytes from 172.18.0.3: seq=0 ttl=64 time=0.328 ms
64 bytes from 172.18.0.3: seq=1 ttl=64 time=0.217 ms

--- caja2 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.217/0.272/0.328 ms
```

4.5 Desconectar un contenedor de la red

```
andrea@Andrea:~/docker-lab$ docker network disconnect mi_red caja2
andrea@Andrea:~/docker-lab$
```

Comprobación:

```
andrea@Andrea:~/docker-lab$ docker exec caja1 ping -c 2 caja2
ping: bad address 'caja2'
andrea@Andrea:~/docker-lab$
```

5) Docker Compose

```
andrea@Andrea:~/docker-lab$ cat <<EOF > compose.yml
services:
  web:
    image: nginx:latest
    ports:
      - "8082:80"
EOF
andrea@Andrea:~/docker-lab$
```

5.2 Levantar y Probar

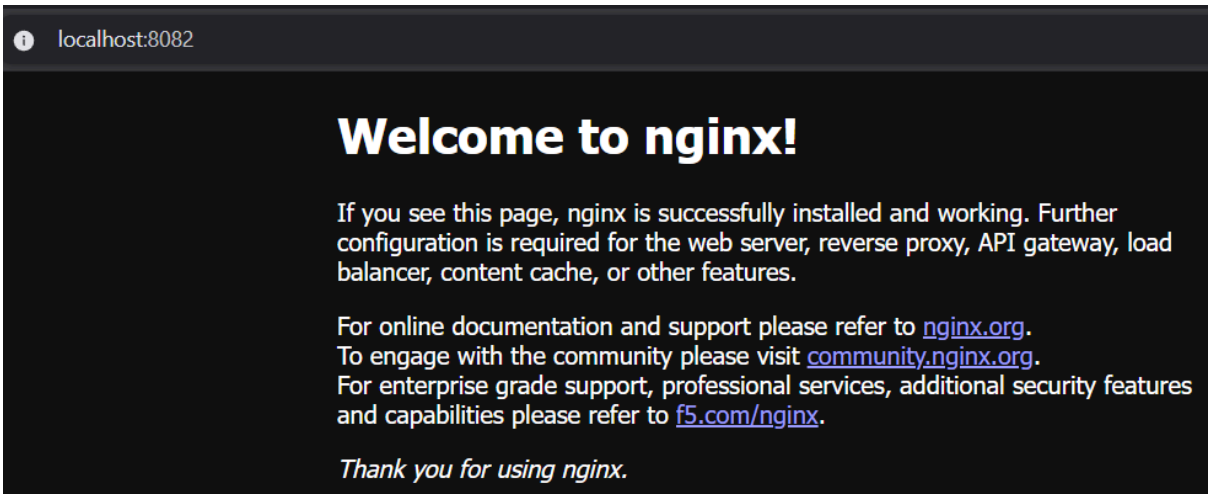
Levantar el servicio:

```
andrea@Andrea:~/docker-lab$ docker compose up -d
[+] up 2/2
 ✔Network docker-lab_default Created
 ✔Container docker-lab-web-1 Started
```

Comprobar:

```
andrea@Andrea:~/docker-lab$ docker compose ps
NAME                IMAGE          COMMAND                  SERVICE   CREATED          STATUS          PORTS
docker-lab-web-1    nginx:latest   "/docker-entrypoint.…"   web       47 seconds ago   Up 46 seconds   0.0.0.0:8082->80/tcp, [::]:8082->80/tcp
```

Prueba de acceso:



5.3 al 5.5 Gestión y Escalamiento

Ver logs:

```
andrea@Andrea:~/docker-lab$ docker compose logs
web-1  | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
web-1  | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
web-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
web-1  | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
web-1  | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
web-1  | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
web-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
web-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
web-1  | /docker-entrypoint.sh: Configuration complete; ready for start up
web-1  | 2026/04/06 03:14:02 [notice] 1#1: using the "epoll" event method
web-1  | 2026/04/06 03:14:02 [notice] 1#1: nginx/1.29.7
web-1  | 2026/04/06 03:14:02 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
web-1  | 2026/04/06 03:14:02 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
web-1  | 2026/04/06 03:14:02 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
web-1  | 2026/04/06 03:14:02 [notice] 1#1: start worker processes
web-1  | 2026/04/06 03:14:02 [notice] 1#1: start worker process 29
web-1  | 2026/04/06 03:14:02 [notice] 1#1: start worker process 30
web-1  | 2026/04/06 03:14:02 [notice] 1#1: start worker process 31
web-1  | 2026/04/06 03:14:02 [notice] 1#1: start worker process 32
web-1  | 2026/04/06 03:14:02 [notice] 1#1: start worker process 33
web-1  | 2026/04/06 03:14:02 [notice] 1#1: start worker process 34
web-1  | 2026/04/06 03:14:02 [notice] 1#1: start worker process 35
web-1  | 2026/04/06 03:14:02 [notice] 1#1: start worker process 36
web-1  | 2026/04/06 03:14:02 [notice] 1#1: start worker process 37
web-1  | 2026/04/06 03:14:02 [notice] 1#1: start worker process 38
web-1  | 2026/04/06 03:14:02 [notice] 1#1: start worker process 39
web-1  | 2026/04/06 03:14:02 [notice] 1#1: start worker process 40
web-1  | 172.19.0.1 - - [06/Apr/2026:03:15:40 +0000] "GET / HTTP/1.1" 200 896 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome
/146.0.0.0 Safari/537.36" "-"
```

Escalar (Simulación de Alta Disponibilidad):

```
andrea@Andrea:~/docker-lab$ docker compose up -d --scale web=3
[+] up 2/3
 ✔Container docker-lab-web-1 Running                                                                                    0.0s
 ✔Container docker-lab-web-3 Created                                                                                    0.2s
 ⠼ Container docker-lab-web-2 Starting                                                                                   0.6s
Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint docker-lab-web-2 (230e53ca542c3ea586c0d0413
1d754cc03bc7cb96eff5363f9ea26e5c0cf282c): Bind for 0.0.0.0:8082 failed: port is already allocated
andrea@Andrea:~/docker-lab$
```

6) Dockerfile: probar instrucciones principales

6.1 Construcción con Argumentos

Creación de carpeta, creación del archivo de texto y creación del Dockerfile "maestro":

```
andrea@Andrea:~/docker-lab$ mkdir ../dockerfile-test && cd ../dockerfile-test
andrea@Andrea:~/dockerfile-test$ echo "contenido local" > archivo.txt
andrea@Andrea:~/dockerfile-test$ cat <<EOF > Dockerfile
FROM alpine:3.20
LABEL autor="prueba"
ARG VERSION=1.0
ENV APP_ENV=dev
WORKDIR /app
COPY archivo.txt /app/archivo.txt
RUN echo "version=\$VERSION" > /app/version.txt
ADD archivo.txt /app/archivo2.txt
VOLUME ["/data"]
CMD ["sh", "-c", "echo APP_ENV=\$APP_ENV && cat /app/archivo.txt && cat /app/version.txt"]
EOF
andrea@Andrea:~/dockerfile-test$
```

Construcción de la imagen usando un argumento:

```
andrea@Andrea:~/dockerfile-test$ docker build -t dockerfile-demo --build-arg VERSION=2.0 .
[+] Building 2.1s (10/10) FINISHED                                                                                  docker:default
 => [internal] load build definition from Dockerfile                                                                          0.1s
 => => transferring dockerfile: 344B                                                                                          0.0s
 => [internal] load metadata for docker.io/library/alpine:3.20                                                                0.1s
 => [internal] load .dockerignore                                                                                             0.1s
 => => transferring context: 2B                                                                                               0.0s
 => [internal] load build context                                                                                             0.1s
 => => transferring context: 54B                                                                                              0.0s
 => [1/5] FROM docker.io/library/alpine:3.20@sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03805          0.1s
 => => resolve docker.io/library/alpine:3.20@sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03805          0.1s
 => CACHED [2/5] WORKDIR /app                                                                                                 0.0s
 => [3/5] COPY archivo.txt /app/archivo.txt                                                                                   0.1s
 => [4/5] RUN echo "version=2.0" > /app/version.txt                                                                           0.6s
 => [5/5] ADD archivo.txt /app/archivo2.txt                                                                                   0.1s
 => exporting to image                                                                                                        0.9s
 => => exporting layers                                                                                                       0.4s
 => => exporting manifest sha256:2e22080be1e7807e96aab37538e8816ce10f6b4b1cc046c846a87b3f93e0f5cf                             0.1s
 => => exporting config sha256:82c3a94a5f153c21ddf489af2327eadb451e0e6577635eb0085dc973802ab1bd                               0.0s
 => => exporting attestation manifest sha256:02b2d49b791f093cc180ebbed3e687e42ac28bbe54ccaf7e04cebcc23aa96f90                 0.2s
 => => exporting manifest list sha256:06be91f54dcbaed997a09cd385cca63df2cd68e88f035f042d4ce892981ff469                        0.0s
 => => naming to docker.io/library/dockerfile-demo:latest                                                                     0.0s
 => => unpacking to docker.io/library/dockerfile-demo:latest                                                                  0.1s
andrea@Andrea:~/dockerfile-test$
```

Ejecución y comprobación del resultado:

```
andrea@Andrea:~/dockerfile-test$ docker run --rm dockerfile-demo
APP_ENV=dev
contenido local
version=2.0
andrea@Andrea:~/dockerfile-test$
```

6.2 Inspeccionar label, env y volume

```
andrea@Andrea:~/dockerfile-test$ docker image inspect dockerfile-demo
[
    {
        "Id": "sha256:06be91f54dcbaed997a09cd385cca63df2cd68e88f035f042d4ce892981ff469",
        "RepoTags": [
            "dockerfile-demo:latest"
        ],
        "RepoDigests": [
            "dockerfile-demo@sha256:06be91f54dcbaed997a09cd385cca63df2cd68e88f035f042d4ce892981ff469"
        ],
        "Comment": "buildkit.dockerfile.v0",
        "Created": "2026-04-06T03:25:05.534384405Z",
        "Config": {
            "Env": [
                "PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
                "APP_ENV=dev"
            ],
            "Cmd": [
                "sh",
                "-c",
                "echo APP_ENV=$APP_ENV && cat /app/archivo.txt && cat /app/version.txt"
            ],
            "Volumes": {
                "/data": {}
            },
            "WorkingDir": "/app",
            "Labels": {
                "autor": "prueba"
            },
            "ArgsEscaped": true
        },
        "Architecture": "amd64",
        "Os": "linux",
        "Size": 3632914,
        "RootFS": {
            "Type": "layers",
            "Layers": [
                "sha256:f5b2c0456bf1080bb3f0617295df9721aa4aa27a536bba8e301c3af8d530c9cb",
                "sha256:b02b5695ec2f0937edde918d4d50a4f89fe42458e1fe23149fceccfcf40afb56",
                "sha256:c66621c4573de882ffc5652ecafd39b77cac561015a416911f44b7441001e016",
                "sha256:766ac7c894992a05b64ebc8db8c8ce3b5728ac42853a3264a1b151a889d9d030",
                "sha256:fcd2f1b6c61c2e94fb65bd5100f5e2b6acfaf286d94f62f4bcda4f17d6a82c74"
            ]
        },
        "Metadata": {
            "LastTagTime": "2026-04-06T03:25:06.26801713Z"
        },
        "Descriptor": {
            "mediaType": "application/vnd.oci.image.index.v1+json",
            "digest": "sha256:06be91f54dcbaed997a09cd385cca63df2cd68e88f035f042d4ce892981ff469",
            "size": 856
        },
        "Identity": {
            "Build": [
                {
                    "Ref": "hzoeor17x72p5un4b4gj9ifew",
                    "CreatedAt": "2026-04-06T03:25:06.380593611Z"
                }
            ]
        }
    }
]
andrea@Andrea:~/dockerfile-test$
```

Comprobaciones puntuales:

```
andrea@Andrea:~/dockerfile-test$ docker image inspect dockerfile-demo --format '{{json
.Config.Labels}}'
docker image inspect dockerfile-demo --format '{{json .Config.Env}}'
docker image inspect dockerfile-demo --format '{{json
.Config.Volumes}}'
{"autor":"prueba"}
["PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin","APP_ENV=dev"]
{"/data":{}}
andrea@Andrea:~/dockerfile-test$
```